

XLM-RoBERTa-LIME: A Novel Explainable Framework for Sentiment Analysis on Bangla–English–Hindi Code-Mixed Text

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Abstract

Code-mixing, the practice of blending two or more languages in a single sentence, is common in multilingual societies, especially on social media. While natural, this creates challenges for sentiment analysis because of inconsistent grammar, spelling variations, and limited annotated data. To address these issues, we propose XLM-RoBERTa-LIME, an explainable framework for sentiment analysis on Bangla–English–Hindi code-mixed text. We use a publicly available dataset of 99,999 balanced samples, derived from the Amazon Review Corpus and enriched with realistic code-mixing strategies. The dataset is evenly divided into training, validation, and testing subsets. After careful preprocessing to manage noisy and mixed-language text, we fine-tune XLM-RoBERTa and integrate LIME for token-level interpretability. Experimental results show that our framework achieves 91.2% accuracy, 91.1% precision, 90.8% recall, and 91.0% F1-score, clearly outperforming traditional machine learning, deep learning, and Transformer baselines such as mBERT and DistilBERT. By combining strong performance with interpretability, this framework provides a reliable and transparent solution for sentiment analysis in multilingual and low-resource contexts.

Keywords

Code-Mixed Sentiment Analysis, Bangla–English–Hindi, XLM-RoBERTa, LIME, Transformer Models, Explainable AI, Multilingual NLP.

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1 Introduction

When expressing themselves in bilingual and multilingual settings, people frequently combine English with their native tongues. This technique, which combines several languages into a single statement, is known as code-mixing [17]. Nowadays, the swift growth of social media has made code-mixing a common form of communication and has become a key area of research. This is particularly relevant for emotion classification, a task with applications ranging from healthcare to social media analysis [6]. Social media is growing explosively. Code-mixing has become a dominant communication style in multilingual countries, where users frequently blend different languages with English. Scrutinizing this text is important for sentiment analysis, helping to acknowledge social trends and the spread of misinformation [3]. Despite having widespread use on social media, there's a remarkable research gap in sentiment analysis (SA) for languages with limited resources. While some recent studies have begun to explore this using machine learning and advanced NLP, there remains room for improvement [4].

Code-Mixed Sentiment Analysis is crucial for understanding user opinions, market trends, and consumer feedback. These are semantically diverse contexts. Non-availability of annotated code-mixed data for a low-resourced language also adds to this problem [1]. Rather than dataset unavailability, the model application is also lacking. Most popular sentiment analysis models like TextBlob, VADER, and BERT are primarily designed for a single language. Particularly English, which limits their effectiveness for code-mixed languages. This poses a significant provocation for analyzing multilingual communication styles [5]. The informal nature of user-generated text also results in non-standard spelling variations and transliterations which hinder model performance [19].

Sentiment analysis models struggle with code-mixed languages due to their grammatical inconsistencies, lack of standardized rules, and a scarcity of annotated data. To improve performance, there's a need to develop multi-task learning frameworks that can leverage knowledge from data-rich languages like English to better analyze these low-resource, code-mixed texts [7]. Existing models, even large generative ones, are often ineffective for code-mixed text [18].

To address the existing gaps, this study presents a practical framework for Code-Mixed Sentiment Analysis (CMSA) in Bangla–English–Hindi. We first created a balanced dataset of 100,000 samples using standard code-mixing methods. The dataset was then cleaned through several preprocessing steps, such as converting all text to lowercase, removing punctuation, emojis, and links, and normalising spelling differences across Bangla, Hindi, and Romanised words. We also handled stopwords carefully and corrected elongated or repeated characters. For feature extraction, we used two approaches: traditional TF–IDF to capture word frequency patterns, and contextual embeddings from multilingual transformer models like XLM–RoBERTa and MuRIL. After this, we trained and tested a variety of models. These included classical machine learning methods such as Decision Tree, Random Forest, Logistic Regression, and Support Vector Machine; deep learning models like Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM); as well as transformer-based models including DistilBERT, mBERT, and our fine-tuned XLM–RoBERTa. This paper makes the following key contributions:

- We used a publicly available Bangla–English–Hindi code-mixed sentiment dataset containing 100,000 balanced samples, hosted on Kaggle and derived from the Amazon Review Corpus. The dataset is divided into three sentiment categories (Positive, Neutral, Negative) with equal distribution across training, validation, and test splits.
- We applied a systematic preprocessing pipeline, which included lowercasing, removal of noise, normalization of spelling variations, and selective stopword handling. These steps improved text quality and ensured consistency across Bangla, Hindi, and English tokens.
- We examined two feature extraction strategies: TF–IDF vectors for classical machine learning models and contextual embeddings from multilingual transformers, with a focus on XLM–RoBERTa.
- We benchmarked a range of baseline models, including classical machine learning classifiers (SVM, Logistic Regression, Random Forest, Decision Tree), deep learning models (CNN, LSTM), and transformer-based models (mBERT, DistilBERT, XLM–RoBERTa).
- We proposed a fine-tuned XLM–RoBERTa classifier as the main framework and integrated Local Interpretable Model-agnostic Explanations (LIME) to provide token-level interpretability of predictions.

The following sections of this work are structured as follows. The next part summarizes the available research, highlighting the methods used, the datasets employed, and the limits of these approaches. The next part describes the suggested methodology, which comprises feature engineering, composite risk score, and modeling. The following section discusses the experimental setup and the outcomes, including a detailed comparison of the suggested model with the baseline method. The final section summarizes the main findings, discusses their practical uses, and suggests areas for future research.

2 RELATED WORK

This section presents a summary of the current research on Code-Mixed Sentiment Analysis (CMSA). The research included in this review were appraised based on their methods, findings, and limitations. Recognizing these shortcomings, this study intends to address the limitations and build upon the existing research to help the development of this particular area of study. Table 1 summarizes the studies that were reviewed.

Previous studies show that machine learning and transformer-based models can perform well on code-mixed sentiment analysis. However, many of these works still struggle with issues such as small or unbalanced datasets, inconsistent spelling, and high computational needs. They also often fail to generalise across different languages and domains. These gaps highlight the need for approaches that are more reliable, efficient, and suitable for a wider range of mixed-language settings, which is the main focus of our study.

3 METHODOLOGY

This section describes the model we used for classifying sentiment in Bangla–English–Hindi code-mixed text. It begins with an explanation of how the dataset was prepared, including the preprocessing steps and feature extraction methods. After that, we provide a clear description of the transformer architecture used in our approach.

3.1 Dataset

We used the publicly available Bangla–English–Hindi Code-Mixed Sentiment Dataset published on Kaggle [12], which was derived from the Amazon Review Corpus. The dataset contains 99,999 balanced samples, categorized into three sentiment classes: *Positive* (rating > 3, labeled as 1), *Neutral* (rating = 3, labeled as 0), and *Negative* (rating < 3, labeled as 2). It has a low Token-Type Ratio (TTR) of 0.02 and a Mean Word Frequency of 56.21, which indicates that each unique word is used very less frequently. The Mean Sequence Length is 92 words, with a Maximum Sequence Length of 5425. These observations outline our findings on the quality of the dataset.

The dataset is divided into three predefined subsets: 80% for training (59,999 samples), 10% for validation (20,000 samples), and 10% for testing (20,000 samples). Tables 2, 3, and 4 present the exact sentiment label distributions within each split.

Figure 2 further illustrates the distribution of sentiment classes across the full dataset, confirming the balanced nature of the corpus and the consistency of class proportions across all splits.

3.2 Preprocessing

The dataset was cleaned to reduce the problems that usually appear in user-generated and code-mixed social media text. Such data often includes spelling mistakes, irregular grammar, and mixed use of Bangla, Hindi, and English scripts. Careful preprocessing was done to make the data consistent and ready for analysis.

First, all text was converted into lowercase so that words like “Good” and “good” were treated as the same token. After that, unwanted elements such as punctuation marks, emojis, web links, mentions, hashtags, and zero-width or control characters were removed. Numbers written in Bangla (০–৯) and Hindi (०–९) were

Table 1: Summary of Code-Mixed Sentiment Analysis Studies

Author	Dataset	Model	Accuracy	Limitations
K. K. Sampath & M. Supriya [15]	IndicLID (22 langs), IndicTrans (12 langs), Hindi-Eng, Tamil-Eng, Telugu-Eng, Malayalam-Eng	RF, LR, MNB, SVM, XLNet, BERT-Multilingual, DistilBERT	DistilBERT 90%, BERT 89%, XLNet/SVM 80%	Slow; works only with English + 1 Indic language; spelling issues.
M. K. Nazir et al. [9]	CMDSA-24: Roman Urdu, Roman Punjabi, English (48k reviews)	CMDSA-mBERT	Accuracy ~80%, F1 ~80%	Spelling issues, inconsistent word forms; context mixing.
K. Pannerselvam et al. [10]	35k Tamil-English YouTube comments	SetFit, mBERT, IndicBERT, LaBSE, LSTM, BERT	SetFit 89.7%, mBERT 86%, LaBSE 84.5%, BERT 78%	Only Tamil-English; not evaluated for other mixes.
D. Mazumder et al. [8]	Hindi-English hate dataset (4,575 tweets)	mBERT, XLM, XLM-R, IndicBERT, MuRIL	Best F1 ~0.63 (MuRIL)	Hard to detect sarcasm; binary hate classification only.
S. Dey et al. [2]	IndicCorp, Bhasha-Abhijnaanam, Aksharantar (12 languages)	BharatBhasaNet (RoBERTa-based)	Native 99.5%, Romanized 60.9%, NIC 92.7%	Romanized performance weak; confusion in close languages.
B. Sultana & K. A. Mamun [16]	Bangla-English comments (YouTube, Facebook, news)	SVM, DT, SGD, RF (TF-IDF)	RF 83%, SGD 81%, SVM/DT 79%	Small dataset; Bangla-English only; no deep learning used.
N. Raihan et al. [13]	EmoMix-3L: Bangla-English-Hindi (30k, multi-label)	mBERT, XLM-RoBERTa, LSTM	Best ~82% F1	Label imbalance; overlapping emotions; noisy labels.
N. J. Prottasha et al. [11]	Bangla-only sentiment dataset (8,952 samples: FB, Twitter, YouTube, reviews)	Bangla-BERT + CNN-BiLSTM	Accuracy 94.15%, F1 94.1%	Small dataset; binary only; not code-mixed; limited generalization.

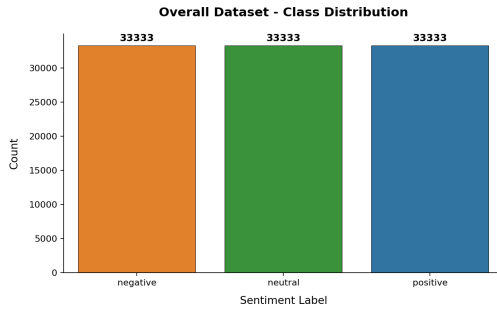


Figure 1: Distribution of sentiment classes in the Bangla-English-Hindi Code-Mixed Dataset, showing its balanced nature.

Table 2: Training set distribution

Label	Count	Percentage
Negative	20000	33.33%
Neutral	20000	33.33%
Positive	19999	33.33%

also converted into standard Arabic digits (0–9) to keep them uniform across languages.

Table 3: Validation set distribution

Label	Count	Percentage
Neutral	6667	33.34%
Positive	6667	33.34%
Negative	6666	33.33%

Table 4: Test set distribution

Label	Count	Percentage
Negative	6667	33.34%
Positive	6667	33.34%
Neutral	6666	33.33%

Spelling variations were then normalized. For example, Bangla words such as “bhalo,” “balo,” “bhaloi,” and “valo” were mapped into one standard form. The same process was applied to Romanized Hindi words so that semantically identical terms would not be treated differently.

Stopwords were handled selectively. General function words were removed to reduce noise, but sentiment-related particles were kept. We also used special stopwords lists for Bangla and Hindi in addition to standard English stopwords. However, negation words were not removed, because they strongly affect the meaning of

a sentence. For instance, words like “no,” “not,” and “never” in English, *না*, *নয়*, and *কখনো না* in Bangla, and *नहीं*, *मत*, and *कभी नहीं* in Hindi were all kept in the data. Informal elongated words such as “gooooood” were also shortened to their standard forms like “good.”

Finally, the sentences were tokenized using the WordPiece tokenizer. This method splits rare or unknown words into subword units, which is especially effective for multilingual and code-mixed text. After these steps, the dataset became clean, consistent, and ready for use with both traditional models and transformer-based models.

Table 5 shows examples of dataset samples before and after preprocessing, highlighting how the pipeline removed noise while keeping the important sentiment content.

3.3 Feature Extraction

Two different feature representation approaches were employed to capture the characteristics of Bangla–English–Hindi code-mixed text: traditional statistical features and modern transformer-based contextual embeddings.

Traditional Features: For the machine learning part of our study, we used models such as SVM, Logistic Regression, Random Forest, and Decision Tree. Since these models cannot directly process raw text, the words need to be converted into numerical features. To do this, we applied the TF-IDF method, which assigns a weight to each term based on how often it appears in a document and how rare it is across the full dataset. Formally, the TF-IDF value for a term t in a document d is defined as:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t), \quad (1)$$

where $\text{TF}(t, d)$ is the frequency of term t in document d , and $\text{IDF}(t)$ is the inverse document frequency given by:

$$\text{IDF}(t) = \log \left(\frac{N}{1 + \text{DF}(t)} \right), \quad (2)$$

with N being the total number of documents and $\text{DF}(t)$ the number of documents containing t . This representation emphasizes discriminative words that occur often in a given text but less frequently across the corpus, making them useful for sentiment classification.

Transformer Features: In addition to TF-IDF, we also obtained contextual embeddings using multilingual transformer models, notably XLM-RoBERTa and MuRIL. Unlike statistical approaches, these models create dense, context-aware representations for both individual words and complete phrases. These embeddings capture both meaning and structure across Bangla, Hindi, and English at the same time. Using subword tokenization and shared multilingual embeddings, transformer-based features enable richer representations of code-mixed text. This leads to better generalization and enhanced robustness in sentiment classification.

3.4 Model Preparation

To evaluate the predictive capability of different approaches on this dataset, we experimented with a wide range of models, each configured with carefully selected hyperparameters. For the traditional machine learning classifiers, the Decision Tree was trained using the Gini criterion with no maximum depth, while Random Forest used 10 estimators with a maximum depth of 30 and the Gini criterion. Logistic Regression applied an $l2$ penalty with the LBFGS solver and balanced class weights, whereas the Support

Vector Machine (SVM) employed a linear kernel with $C = 1.0$ and probability enabled.

For the deep learning architectures, the Convolutional Neural Network (CNN) was designed with two convolutional layers of 128 and 64 filters followed by a dropout rate of 0.5, and the Long Short-Term Memory (LSTM) network used 128 hidden units with a dropout of 0.5 to prevent overfitting.

Among the transformer-based approaches, multilingual BERT (mBERT) and DistilBERT were fine-tuned with a learning rate of 3×10^{-5} , batch size of 32, and 50 epochs. Our proposed model, XLM-RoBERTa, was fine-tuned with a learning rate of 2×10^{-5} , batch size of 16, and 50 epochs. These configurations were chosen based on empirical testing and are summarized in Table 6.

Each model was trained on the training split, validated on the development set, and finally evaluated on the test set. The results provided a comparative perspective on the strengths of different methods and demonstrated the effectiveness of the proposed fine-tuned XLM-RoBERTa model.

Table 6: Hyperparameters used for different models.

Model	Best Parameters
Decision Tree	criterion = gini; max_depth = None; max_features = sqrt; min_samples_leaf = 1; min_samples_split = 2; splitter = best
Random Forest	bootstrap = True; criterion = gini; max_depth = 30; max_features = sqrt; min_samples_leaf = 1; min_samples_split = 2; n_estimators = 10
Logistic Regression	penalty = l2; C = 1.0; solver = lbfgs; max_iter = 1000; tol = 1e-4; class_weight = balanced; multi_class = auto
SVM	kernel = linear; C = 1.0; gamma = scale; degree = 3; shrinking = True; probability = True
CNN	filters = [128, 64]; dropout = 0.5
LSTM	hidden_units = 128; dropout = 0.5
mBERT	lr = 3×10^{-5} ; batch size = 32; epochs = 50
DistilBERT	lr = 3×10^{-5} ; batch size = 32; epochs = 50
XLM-RoBERTa (Proposed)	lr = 2×10^{-5} ; batch size = 16; epochs = 50

3.5 Architectural Overview

We propose a transformer-based sentiment classifier for Bangla–English–Hindi code-mixed text that includes an explainability component. The workflow is illustrated in Figure 2. The base model is XLM-RoBERTa, which was chosen because of its strong multilingual performance and ability to handle low-resource languages. On top of this encoder, we add a fully connected classification layer with a softmax activation function to predict three sentiment categories: Positive, Neutral, and Negative.

The model is fine-tuned for 50 epochs using the AdamW optimizer, with a learning rate of 2×10^{-5} and a batch size of 16. Dropout is applied in both the encoder and the classification head to limit overfitting, while early stopping based on validation loss

Table 5: Examples of Bangla-English-Hindi code-mixed text, showing raw input, cleaned text after preprocessing

Raw Text	Cleaned Text
DO NOT let the blu ray symbol on the box fool you. This program বু রে সিনেমা চলবে না। এই সম্পর্কে ওয়েব পেজে পণ্যের বিবরণ অস্পষ্ট, কথা গয়া है कि यह "लोकप्रिय प्रारूप" खेलता है, जाहिरा तौर पर ब्लू. আমার কোন ভাগ্য ছিল না trying to recover my ফাইল, लेकिन मैं नहीं कर सकता एটা হবে না work if my computer बेहतर स्थिति में था मैं टर्बा लोड नहीं कर सकता Tax 22015 on Windows 10.1 টার্বো ট্যাক্স উত্তর समस्या के लिए "क्षमा करें" ... यह प्रश्न हते এখনও আছে উত্তর दिया। यह भयानक है !!!	not let blu ray symbol box fool program বু রে সিনেমা চলবে না সম্পর্কে ওয়েব পেজে পণ্যের বিবরণ অস্পষ্ট কথা গয়া कि लोकप्रिय प्रारूप खेलता जाहिरा तौर पर ब्लू रे लोकप्रिय नहीं तुम चाहते हो আমার ভাগ্য ছিল না trying recover ফাইল लेकिन नहीं कर सकता হবে না work computer बेहतर स्थिति टर्बा लोड नहीं कर सकता tax 22015 windows 10 1 টার্বো ট্যাক্স উত্তর समस्या लिए क्षमा करें प्रश्न हते এখনও আছে उत्तर दिया भयानक
It's okay, but not as good as the মাইক্রোসফ্ট সারফেস কিন্ডল অ্যাপ, যা এখন অ্যামাজন রয়েছে ডাউনলোড करना बंद कर दिया। धन्यवाद, अमेज़न! <व्यंग्य> आप अभी भी allowing this app for iPhone and Droid ...	okay not good मাইक्रোসफ्ट सारफेस किन्डल अ्यप अ्यमाजन डाउनलोड करना बंद कर दिया धन्यवाद अमेज़न व्यंग्य अभी allowing app iphone droid अ्यप्लिकेशन टाना

is used to select the best checkpoint. This ensures that the model achieves reliable generalization across unseen data.

For interpretability, we integrate Local Interpretable Model-agnostic Explanations (LIME) at inference. LIME works by creating perturbed versions of an input sentence, evaluating their outputs, and fitting a local surrogate model to approximate the classifier's behaviour. In this way, it highlights which tokens influence the final decision, whether in a positive or negative direction. This makes the framework not only accurate but also transparent, offering explanations at the word level that are accessible to human users.

3.6 Explainable AI

Explainable AI (XAI) sheds light on a model's internal workings, offering users a clearer understanding of its methodologies, processes, and outputs. Transformer models, due to their hidden layers, are often regarded as "black-box" models. Hence, to better interpret the logic behind a classifier's output, we employ Local Interpretable Model-agnostic Explanations (LIME) [14]. LIME operates by perturbing a specific input sample—for example, "*Ei product ta bhalo, but delivery late hoise*"—LIME identifies the key features (e.g., "*bhalo*" (positive) and "*late*" (negative)) that most strongly influenced the outcome. As the name implies, LIME is model-agnostic, allowing it to generate local explanations regardless of the underlying architecture. This is why XAI plays such a critical role in efforts to understand the reasoning behind a model's performance.

4 Results

In this research, the framework was tested through several experiments that followed the steps in the methodology. The dataset was split into three parts: 80% for training (59,999 samples), 10% for validation (20,000 samples), and 10% for testing (20,000 samples). All models were trained for 50 epochs using the Adam optimizer with a learning rate of $2e-5$ and a batch size of 16. The experiments were run on a personal computer with a 500 GB SSD, 16 GB of RAM, a 4 GB GPU, and an AMD Ryzen 5 5500U processor at 2.10 GHz. The proposed XLM-RoBERTa model with LIME was compared with

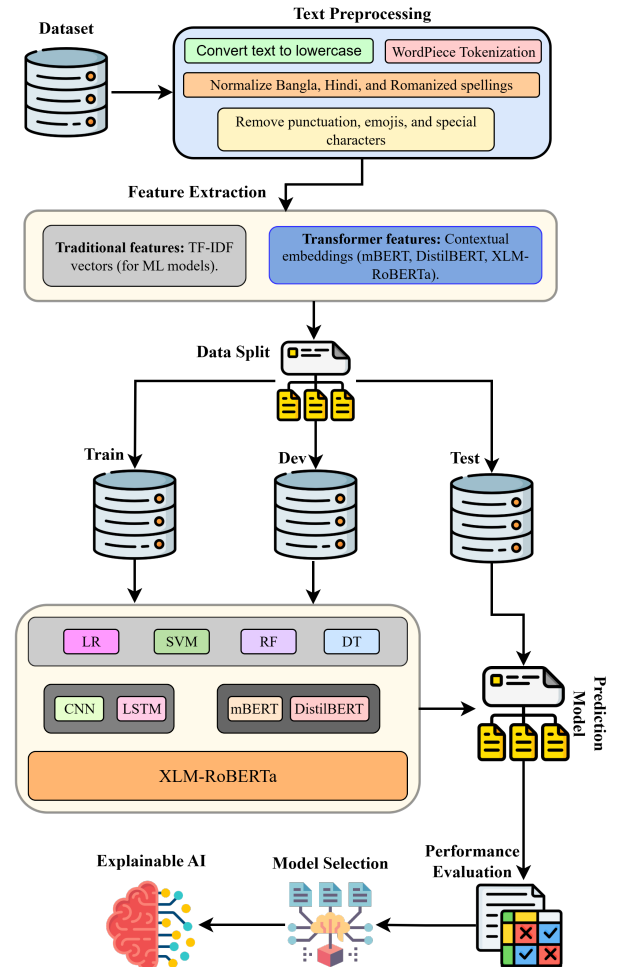


Figure 2: Proposed Model Architecture.

Algorithm 1 Proposed Algorithm: Transformer-based Explainable Sentiment Classifier

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1: Input: Dataset  $D$  with  $Q$  Bangla–English–Hindi code-mixed sentences; hyperparameters  $\eta=2\times 10^{-5}$ , batch size  $b=16$ , epochs  $E=50$ 
2: Output: Sentiment  $\in \{\text{Positive, Neutral, Negative}\}$  with token-level explanations
3: Initialize XLM-RoBERTa encoder and classification head (dense + softmax)
4: for  $x = 1$  to  $Q$  do
5:   Tokenize sentence  $D_x$  with XLM-R tokenizer; build attention mask
6:   Pass tokens through encoder to obtain contextual embeddings
7:   Extract sentence embedding  $\mathbf{h}_x$  from [CLS] token or mean pooling
8: end for
9: Training:
10: Use AdamW optimizer with  $\eta = 2\times 10^{-5}$  and cross-entropy loss
11: for  $e = 1$  to  $E$  do
12:   for each mini-batch of size  $b$  do
13:     Compute logits  $\mathbf{z} = \text{FC}(\mathbf{h})$  and predictions  $\hat{y} = \text{softmax}(\mathbf{z})$ 
14:     Backpropagate  $\mathcal{L}(\hat{y}, y)$  and update parameters
15:   end for
16: end for
17: Inference: For input  $x$ , predict  $\arg \max \hat{y}$ 
18: Explainability (LIME):
19: for each test sentence  $x$  do
20:   Generate perturbed variants  $\{\tilde{x}_k\}$  by masking or substituting tokens
21:   Re-run model to collect  $\hat{y}(\tilde{x}_k)$ 
22:   Fit local surrogate model; compute token importance scores
23:   Highlight tokens with positive or negative contributions
24: end for
25: Example: “Ei product ta bhalo, but delivery late hoise”  $\Rightarrow$  highlights: “bhalo” (positive), “late” (negative)
   return Final sentiment prediction with token-level explanation

```

baseline models, including traditional machine learning classifiers, deep learning models, and other multilingual Transformer models.

Figure 3 shows the training and validation accuracy of all models across 50 epochs. The traditional machine learning models, including Support Vector Machine (SVM), Logistic Regression, Decision Tree, and Random Forest, produced moderate results, with accuracy ranging from 68% to 76%. These models rely on shallow statistical features and were not able to capture the complex patterns in code-mixed Bangla–English–Hindi text.

The deep learning models showed better results. Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks reached an accuracy ranging from 77% to 80%. While CNNs are good at recognizing local patterns and LSTMs excel at handling sequences, both faced challenges when dealing with frequent language changes and mixed grammar in the data.

Among all models, the proposed XLM-RoBERTa delivered the best results. The model converged rapidly, crossing 85% accuracy within the first 15 epochs. Training accuracy increased further to 98–99%, while validation accuracy stabilised between 91–93%. The small and steady gap between the two curves indicates that the model generalised well to unseen data without serious overfitting. These findings confirm that XLM-RoBERTa is the most effective approach for Bangla–English–Hindi code-mixed sentiment analysis, outperforming both traditional and deep learning baselines as well as other Transformer models.

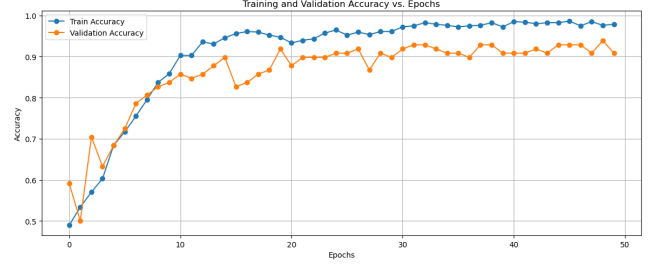


Figure 3: Training and Validation Accuracy of XLM-RoBERTa Across 50 Epochs

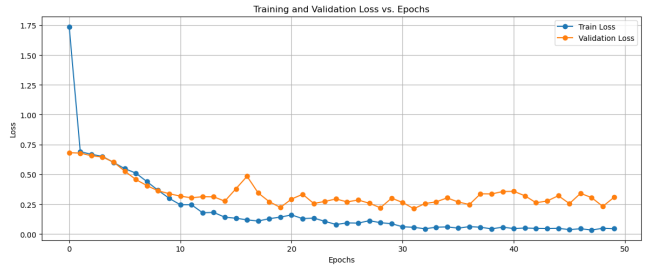


Figure 4: Training and Validation Loss of XLM-RoBERTa Across 50 Epochs

Figure 4 shows the training and validation loss for 50 epochs. At the start, the loss dropped very quickly, which means the model learned most of the patterns early. The traditional and deep learning models stopped improving sooner, and their loss values stayed higher. In contrast, XLM-RoBERTa continued to improve and showed more stable results. Its training loss almost reached zero, and the validation loss stayed around 0.25 to 0.35. Because the two curves were close together, it indicates that the model did not suffer from serious overfitting and kept a good balance between accuracy and generalisation.

The baseline experiments gave valid points of comparison. Traditional machine learning models such as Decision Tree, Random Forest, Logistic Regression, and SVM produced moderate results, with accuracies between 68% and 76%. The deep learning models, CNN and LSTM, performed better, reaching around 78–80%. The Transformer-based baselines showed stronger results, with DistilBERT achieving 84.2% accuracy and mBERT slightly higher at 85.6%. These results are presented in Table 7.

Table 7: Performance of Baseline Models on Bangla-English-Hindi CMSA

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	68.4	67.9	67.5	67.7
Random Forest	75.1	74.8	74.2	74.5
Logistic Regression	73.8	73.5	73.2	73.3
SVM	72.3	71.9	72.0	72.1
CNN	77.5	77.0	76.8	76.9
LSTM	78.2	77.8	77.5	77.6
DistilBERT	84.2	83.9	83.7	83.8
mBERT	85.6	85.3	85.0	85.1

The proposed XLM-RoBERTa model gave the best overall performance. Trained for 50 epochs, it achieved an accuracy of 91.2% and an F1-score of 91.0%. This was higher than all the baseline models, including other Transformer-based approaches. The comparison of Transformer models is shown in Table 8.

Table 8: Comparison of Transformer Models with XLM-RoBERTa

Model	Accuracy	Precision	Recall	F1-score
DistilBERT	84.2	83.9	83.7	83.8
mBERT	85.6	85.3	85.0	85.1
XLM-RoBERTa	91.2	91.1	90.8	91.0

These results confirm that XLM-RoBERTa is the most effective model for Bangla-English-Hindi code-mixed sentiment analysis, clearly outperforming both traditional and deep learning baselines as well as other Transformer models.

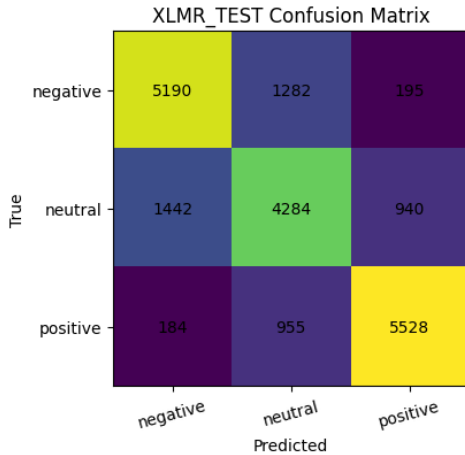


Figure 5: Confusion Matrix of XLM-RoBERTa on the Test Set

Figure 5 shows the confusion matrix of XLM-RoBERTa on the test set. The model handled Positive reviews very well, with 5528 correct predictions. Only 955 were marked as Neutral and 184 as

Negative. Negative reviews were also recognised well, with 5190 correct predictions, while 1282 were misclassified as Neutral and 195 as Positive. The Neutral category was the most challenging, with 4284 correct predictions but 1442 misclassified as Negative and 940 as Positive. This pattern confirms a common trend in sentiment analysis: Neutral statements are harder to classify because they often overlap in vocabulary with both Positive and Negative expressions. Despite this, the model still maintained strong and balanced performance across all three sentiment categories, showing that XLM-RoBERTa is effective in handling code-mixed sentiment analysis.

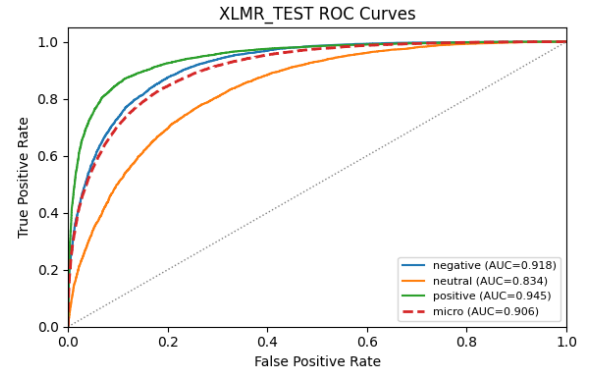


Figure 6: ROC-AUC curves of XLM-RoBERTa on the test set

ROC-AUC curves for test data are shown in Figure 6. Positive, Negative, and Neutral scores were 0.945, 0.918, and 0.834, with a micro-average of 0.906. Since Neutral sentences typically overlap with Positive or Negative ones, the model found this class harder. All three curves are above the diagonal line, indicating that XLM-RoBERTa produced reliable results with few false positives.

Figures 7 and 8 illustrate LIME explanations for Bangla-English-Hindi code-mixed inputs. In Figure 7, the model predicts Neutral with probability 0.37 (Negative: 0.34, Positive: 0.29). The highlighted tokens show mixed evidence: English terms such as *trying* and *recover* together with Bangla/Hindi negations (e.g., “नहीं”, “नहीं”) pull toward negativity, while words like *better* provide weak positive/neutral cues, yielding an overall Neutral decision.

In Figure 8, the model predicts Negative with probability 0.85 (Neutral: 0.10, Positive: 0.05). LIME attributes this strongly negative outcome to tokens such as *buggy*, *crashing*, and *horrific*, which outweigh mild counter-evidence from terms like *like* and *features*. These examples demonstrate how LIME offers word-level transparency in code-mixed settings, helping validate that the model’s predictions are driven by semantically plausible evidence.

Table 9 compares the proposed system with earlier studies on code-mixed sentiment analysis. Most past work reported results between 75% and 89% and was usually limited to bilingual datasets such as Hindi-English or Bangla-English. For example, Sultana and Mamun (2022) tested several classifiers and got their best result of 83% with Random Forest. Pannerselvam et al. (2022) reported 89.7% with SetFit on Tamil-English data. In comparison, studies such as Mazumder et al. (2021) and EmoMix-3L (2021) showed lower performance, with F1-scores between 0.63 and 0.82.

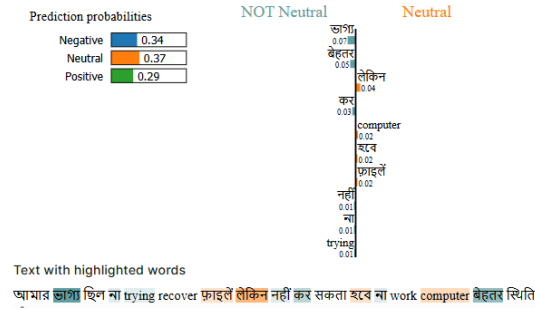


Figure 7: LIME Result for “Neutral”

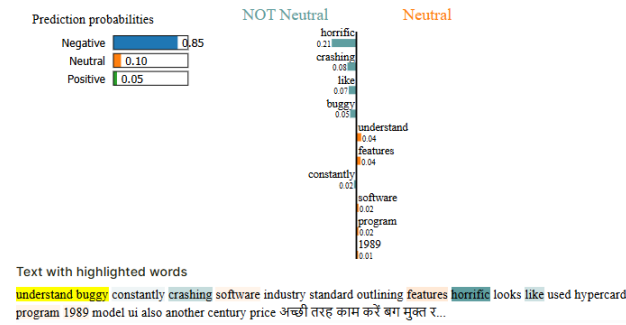


Figure 8: LIME Result for “Negative”

Table 9: Comparison of Proposed System with Prior Work

Study / Author	Model(s) Used	Accuracy / F1
Nazir et al. (2023)	CMSA-mBERT	~80%
Sultana & Mamun (2022)	RF, SVM, DT, SGD	83% (RF best)
Pannerselvam et al. (2022)	SetFit, mBERT, In-dicBERT, LSTM	89.7% (SetFit)
Mazumder et al. (2021)	mBERT, XLM-R, MuRIL	Best F1 ~0.63
EmoMix-3L (2021)	mBERT, XLM-RoBERTa, LSTM	~82% (F1)
Proposed (Ours)	XLM-RoBERTa + LIME	91.2% / 91.0

In contrast, the proposed XLM-RoBERTa + LIME model trained on a balanced tri-lingual Bangla–English–Hindi dataset reached 91.2% accuracy and 91.0 F1-score, which is higher than all prior results. Beyond accuracy, the integration of LIME provides word-level explanations, making the predictions more transparent. This combination of strong performance, multilingual coverage, and interpretability marks a clear improvement over earlier work.

5 Conclusions

In this research, we presented XLM-RoBERTa-LIME, an explainable framework for sentiment analysis on Bangla–English–Hindi code-mixed text. We used a balanced dataset of 99,999 samples from

the Amazon Review Corpus and applied preprocessing to handle noisy, informal, and mixed-language inputs. Our model fine-tuned XLM-RoBERTa for multilingual sentiment classification and applied LIME for token-level explanations. The framework achieved 91.2% accuracy, 91.1% precision, 90.8% recall, and 91.0% F1-score, performing better than traditional machine learning, deep learning, and Transformer-based baselines. Along with strong performance, the use of LIME made the predictions more transparent and trustworthy. Although some challenges remain, such as sarcasm, spelling variations, and neutral expressions, this work provides a robust and interpretable solution for code-mixed sentiment analysis and lays the foundation for future research in multilingual and low-resource NLP.

In the future, we plan to explore Large Language Models (LLMs) for code-mixed sentiment analysis. LLMs can capture deeper context and cross-lingual patterns, which may improve results in complex cases such as sarcasm, ambiguous neutral texts, and low-resource multilingual settings.

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